



FABRIZIO DONATI

Mathematical Modeler | Data Scientist | Machine Learning Expert

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Zürich, Switzerland

STRENGTHS

Mathematical Modeling

Drove innovative solutions by combining problem-solving skills with rigorous analytical thinking and statistical modelling, independently tackling a wide range of technical challenges.

Data Analysis

Optimized data integrity through imputation, anomaly detection, pattern recognition, and rigorous adherence to data quality protocols, enhancing time series analysis in fast-paced settings.

Team Leadership

Led the development of novel features in a variety of cross-departmental projects, providing guidance in data interpretation and management, and directing modeling processes.

SKILLS

C++ · C · C# · Python · MATLAB · SQL ·
.NET · OpenMP · OpenGL · Azure DevOps ·
Git · JIRA · Buck · CMake · Mathematics ·
Statistics · Machine Learning · NLP ·
Parallel Programming ·
Non-Linear Optimization

CERTIFICATIONS

CFA Level I Candidate

CFA Institute (Exam Date: February 2025)
Developing expertise in analyzing financial statements and preparing detailed reports, constructing and managing investment portfolios to optimize returns.

AI for Trading

Udacity
Completed real-world projects designed by industry experts, covering topics from asset management to trading signal generation.

Advanced Statistics and Machine Learning

Coursera

Parallel Computing with CUDA

Pluralsight

PATENTS

Method and system for pressure drop estimation

WO2017158343A1, The Patent Cooperation Treaty (PCT), 2017.

SUMMARY

Proactive modeling expert with a focus on data-driven processes with 10+ years object-oriented programming experience in C++ and Python. Skilled in developing multi-disciplinary simulations, with a strong ability to understand risks and limitations and optimize performance. My methodology includes collaborating with diverse teams, incorporating a deep knowledge of mathematical and statistical concepts into a practical strategy, to ensure that my simulations are not only theoretically sound, but also capable of offering actionable insights.

EXPERIENCE

Senior Simulation Development Engineer

02/2017 - Present

Audi F1 Project

Hinwil, Switzerland

Lead the mathematical modeling and software development of vehicle simulations, leveraging C/C++ for the physical model implementation and C# for the user interface development. Design and deploy state-of-the-art data analysis tools in Python used company-wide.

- Develop software and methods for telemetry data analysis, employing machine learning algorithms to enhance channel information extraction and enable predictive analysis.
- Introduce code performance optimization using parallel algorithms (low- and high- level data and task parallelism in C++) and asynchronous programming to integrate the car model into higher-frequency applications. Optimize system resource utilization, boosting data loading/processing speed by 300% and reducing AWS cloud service costs by 30%.
- Partner with trackside engineers to streamline tools for race preparation and post-event analysis, leading the development of car setup scans and various tools such as lap time, race start, pit lane, ride analysis, wind and strategy simulations.
- Fetch, transform and integrate in Python GPS data, live-sensor measurements and lap time simulations across all race events to fuel regression methodologies for deriving key metrics to rank competitor performances.
- Design the power unit model to explore the impact of the FIA F1 2026 regulation changes, delivering critical simulation insights to the Technical Director.

Software Engineer

01/2023 - 06/2023

Meta

Zürich, Switzerland

Developed software solutions for the high-end mixed reality headset Meta Quest 3.

- Reduced rendering latency in a proprietary virtual reality application by 20%.
- Enhanced the reliability of the system by developing automated end-to-end tests in JavaScript for camera stability, image reconstruction and dynamic range adjustment.
- Redefined a 3D graphics mixed reality occlusion feature in an Android OS application, using OpenGL with C++ for virtual object rotation control.

Computational Biomedical Scientist

01/2012 - 04/2016

King's College London

London, England

Conducted cardiovascular modelling research at St. Thomas' Hospital.

- Improved patient stratification by formulating a novel method to estimate blood pressure from medical imaging data, resulting in a 50% reduction in misdiagnoses.
- Embedded a simplified hemodynamics solver in a multi-physics human heart modeling software, enabling the simulation of complex heart dynamics within clinical timelines.

EDUCATION

Ph.D., Biomedical Engineering

2012 - 2016

King's College London

London, England

- Major: Imaging Sciences

M.Sc., Aerospace Engineering

2009 - 2011

Politecnico di Torino

Turin, Italy

- Major: Computational Fluid Dynamics

B.Sc., Aerospace Engineering

2005 - 2009

Politecnico di Torino

Turin, Italy

- Major: Experimental Aerodynamics

LANGUAGES

English

Native



Italian

Native

